RESULTS BRIEFING

CLIP & VLM anomaly detection

Findings, methodology, testing approach and evaluation

Fast inference did not translate into reliable event detection.

4 + 1

Image models + VadCLIP

3,350

Confirmed inputs / model

13,400

GPU quality predictions

Five target events: smoking, littering, loitering, rough sleeping and unattended items. Only smoking and unattended items currently have usable confirmed cohorts.

ONE-SLIDE SUMMARY

What we learned — and how we tested it

Current-protocol findings; exploratory evidence, with no deployment winner

Methodology

Review event labels; deduplicate inputs; keep recording groups together; fit thresholds on development data.

Testing approach

GPU quality: 13,400 predictions, no pacing. Separate four-core CPU latency pilot. VadCLIP is a supervised study.

Findings

Smoking recall ≤5.41% for current image-model prompts. SmolVLM flags every unattended input; MiniCPM stays normal.

Evaluation limits

Smoking is development-only. Unattended testing is tiny. Three tasks lack verified event labels and matched normals.

Compare model families on shared inputs

Four pretrained image models; VadCLIP uses a different supervised protocol

CLIP similarity

MobileCLIP2-S2
SigLIP2-B/16-224

Continuous anomaly scores. Native image processing and cached text features.

Generative VLMs

SmolVLM-500M-Instruct
MiniCPM-V 4.0

Native chat formatting. Hard labels, raw responses and abstention tracking.

VadCLIP adaptation

Frozen CLIP-B/16 features plus a trained temporal adapter.

Frame supervision and future context; offline smoking study.

Compare image models within a cohort. Keep the supervised VadCLIP results separate.

Only two tasks have usable event labels

Downloaded files do not automatically establish the target anomaly

Target	Local material	Annotation meaning	Scoring readiness
Smoking	3,278 frames	Smoking / Not Smoking; 16 groups	Development only
Unattended items	36 endpoints + 36 windows	Reviewed CAVIAR events / controls	Small dev/test cohort
Littering	6,721 images	Action boxes; rubbish disposal unverified	Needs event / normal review
Loitering	1,675 images 95 videos	Person boxes; no duration-event mapping	Needs event intervals
Rough sleeping	224 files 112 unique	Laying / standing pose proxies	Needs verified sleeping events

Missing tasks remain unscored. No proxy labels or missing annotations are converted into event negatives.

METHODOLOGY

Protect the data before measuring quality

A repeatable pipeline from source annotations to matched model comparisons

01

Review labels

Verify events, normal controls and recording identity.

02

Standardize

Build task-specific manifests and frame/window inputs.

03

Dedup & split

Check pixels and source moments; hold groups together.

04

Run models

Identical ordered inputs, native processors, saved responses.

05

Calibrate & score

Choose dev thresholds; freeze test; bootstrap groups.

Leakage checks + shared cohorts + recorded provenance make the comparison auditable.

Accuracy and latency are separate tests

Use GPU capacity for quality; freeze CPU resources only for the latency pilot

GPU accuracy evaluation

RTX 5000 Ada • 32 GB VRAM
All 3,350 confirmed inputs / model
No pacing sleeps or default RAM cap
CLIP: float32, batch 32
VLMs: bfloat16, one input / call
768-pixel letterbox; 64-token budget

CPU latency pilot

Threadripper PRO 7975WX
Four fixed logical CPUs + four threads
Float32, one worker, batch one
Same eight dev images, three repeats
Three warmups; 24 observations / model
Decode → inference → parse included

GPU timings are observational. CPU timing is a small host-specific replay, excluding capture and queueing.

Measure misses and false alarms together

Accuracy alone can reward a model that always returns normal

Recall

How many true event inputs were detected?

Precision & F1

How reliable are alarms, and how well is recall balanced?

False-positive rate

What fraction of normal inputs were falsely flagged?

AP & AUROC

Continuous score ranking only. No invented VLM AUC.

Coverage

Invalid outputs and abstentions remain visible and count in quality.

Source-group intervals

2,000 paired bootstrap draws over recordings, not isolated frames.

Current prompts miss most smoking positives

3,278 development frames • 1,422 positive • 1,856 normal • 16 known source groups

Model	Precision	Recall	F1	FPR	AP	AUROC
MobileCLIP2-S2	66.67%	0.56%	0.011	0.22%	0.424	0.478
SigLIP2-B/16-224	0.00%	0.00%	0.000	0.00%	0.443	0.512
SmolVLM-500M	40.31%	5.41%	0.095	6.14%	—	—
MiniCPM-V 4.0	0.00%	0.00%	0.000	0.00%	—	—

MobileCLIP detects 8 positives; SmolVLM detects 77 out of 1,422.

Always-normal accuracy is 56.6%. Its smoking recall is zero.

Constant answers do not show event detection

Per test protocol: 4 positive + 12 normal endpoints; only 3 source groups

Model	Frame F1	Window F1	Window recall	Window FPR
MobileCLIP2-S2	0.000	0.250	25.00%	25.00%
SigLIP2-B/16-224	0.000	0.000	0.00%	25.00%
SmolVLM-500M	0.400	0.400	100.00%	100.00%
MiniCPM-V 4.0	0.000	0.000	0.00%	0.00%

SmolVLM: every input is an alarm. MiniCPM: every input is normal. Both patterns fail to distinguish events from controls.

Calibration: MobileCLIP frame recall reaches 100%, but FPR is 66.7%; low-FPR thresholds detect no test positives.

VadCLIP shows limited score discrimination

A separate supervised offline smoking reanalysis; not a pooled zero-shot ranking

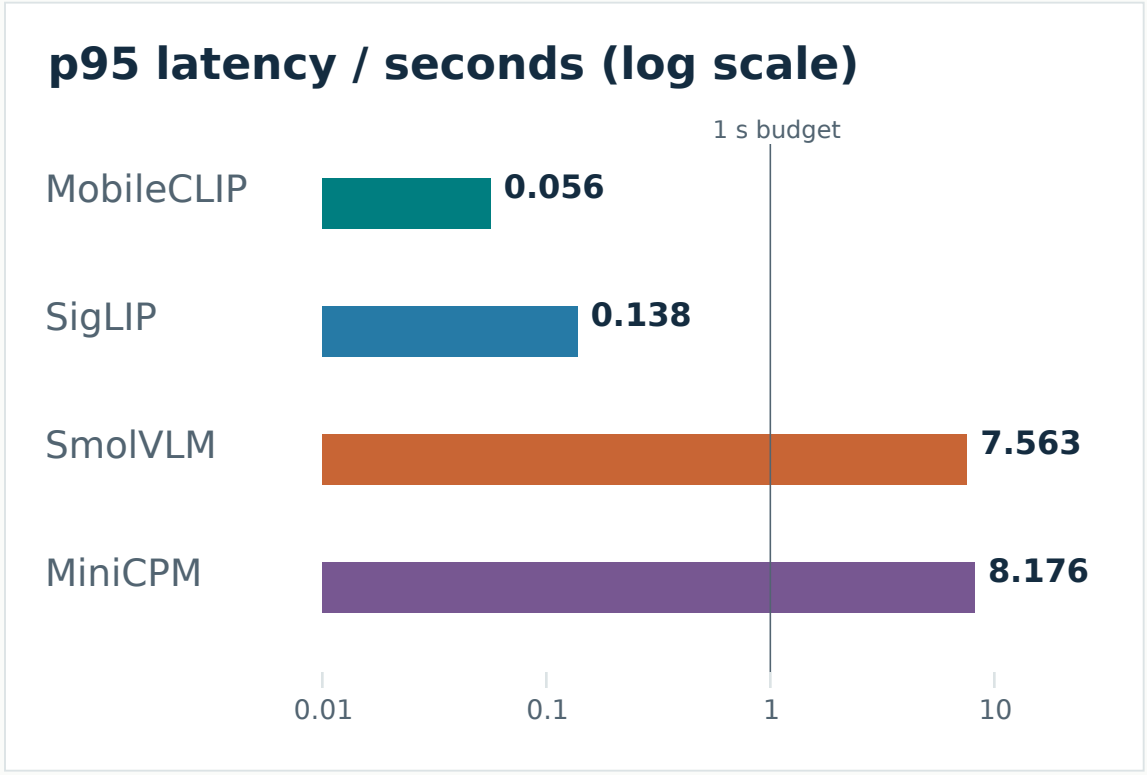
Train	Development	Test
1,910 frames / 9 sessions	312 frames / 4 sessions	1,056 frames / 3 sessions

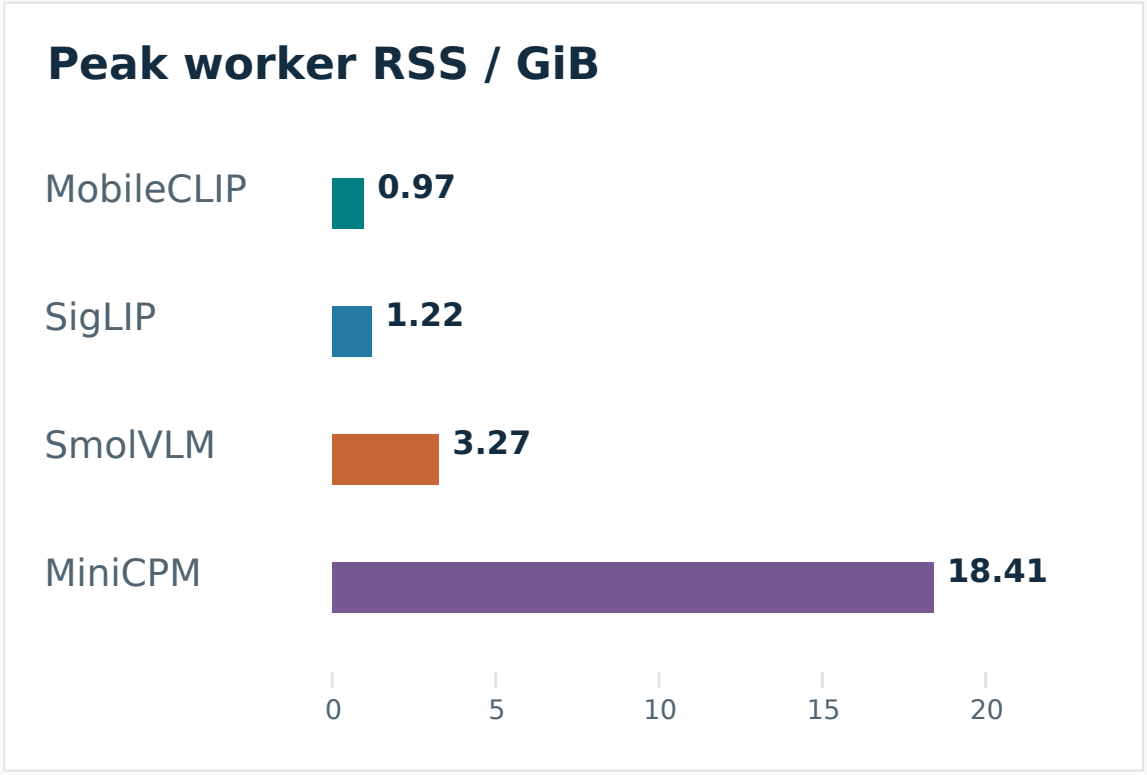
0.633	51.79%	35.69%
Average precision	**Test recall**	**False-positive rate**

AUROC 0.587; F1 0.565. Visual / dual_max branches collapse to all-positive scores. Adapter-only timings exclude the encoder and cannot be compared with end-to-end VLM latency.

CLIP is cheaper in the CPU latency pilot

Same 8 development images × 3 repeats • float32 • 4 logical CPUs • batch one





Both CLIPs met a 1-second inference budget here. The VLMs did not. These costs are host- and protocol-specific.

Remove limits that hid evaluation capacity

Full GPU quality runs; explicit sampling and measurement semantics

Area	Earlier limitation	Updated workflow
Execution	Forced CPU / GPU hidden	Explicit CPU / CUDA placement + dtype
Prepared cohort	Hidden 8-per-label/recording cap	3,350 / 3,350 confirmed inputs retained
Resources	Restrictive defaults	No pacing / default RAM cap; GPU batches
Data lookup	Repeated 3 MB receipt parsing	Indexed once; invalidates on changes
Reporting	Incomplete quality / timing context	AP, recall, FPR, coverage and batch kind

144 tests passed in each environment; all 20 GPU model/cohort summaries succeeded.

No five-task deployment winner is supported

Fix event coverage and test fresh recordings before making a model choice

1 Review event data

Verified positives + task-specific normals.

Recording identity, camera genealogy, duration and ownership rules.

2 Improve the protocol

Test crops and native temporal / image-slicing inputs on development data.

Include detector failures in quality.

3 Freeze & retest

Fresh source recordings; a fixed false-alert budget.

Measure event delay, alerts/hour and real-device latency.

Current findings describe this baseline protocol. They do not exhaust the models' native capabilities.

Prompt sensitivity and uncertainty matter

Development experiments identify hypotheses; they do not validate deployment gains

Smoking model	F1 group CI95
MobileCLIP2-S2	0.002–0.027
SigLIP2-B/16-224	0.000–0.000
SmolVLM-500M	0.006–0.173
MiniCPM-V 4.0	0.000–0.000

Short-prompt diagnostic

MobileCLIP recall: 53.3%
But false-positive rate: 49.4%
AP: 0.460; SigLIP still all normal.

A changed decision bias is not a validated gain.

VadCLIP learning-rate sweep: dev AP 0.369 / 0.393 / 0.425 for 1e-5 / 1e-4 / 1e-3. Selection stayed at 0.001, epoch 1. SmolVLM–MobileCLIP paired F1 interval includes zero (−0.013 to 0.166).

Development thresholds expose the tradeoff

Thresholds were frozen before test; only 16 normal development endpoints per protocol

Input	Model	Dev selection	Threshold	Recall	FPR	F1
frame	MobileCLIP	Max dev F1	0.225	100.00%	66.67%	0.500
frame	MobileCLIP	Dev FPR budget	0.396	0.00%	0.00%	0.000
frame	SigLIP	Max dev F1	-1.926	0.00%	41.67%	0.000
frame	SigLIP	Dev FPR budget	-0.079	0.00%	0.00%	0.000
window	MobileCLIP	Max dev F1	0.409	100.00%	100.00%	0.400
window	MobileCLIP	Dev FPR budget	0.562	0.00%	8.33%	0.000
window	SigLIP	Max dev F1	-1.324	100.00%	100.00%	0.400
window	SigLIP	Dev FPR budget	0.145	0.00%	8.33%	0.000

Small staged source-sequence holdout: no inference about operational alerts/hour or unseen-camera robustness.

Every claim traces back to a saved run

Use the latest combined run; the older IDE report predates these comparisons

Evidence files

Combined GPU report + raw predictions
CPU latency metadata + summaries
Data readiness audit
Workflow review + numeric CSV

Full file paths and sources are in the notes and manifest.

Reproduce & review

Run the isolated GPU helper with:
run --device cuda --per-group 0
--batch-size 32 --vadclip

Freeze the protocol before a fresh test. Inspect false alarms and misses, not only aggregate accuracy.

Evidence is complete for available tasks; verified five-task event coverage remains incomplete.